DECISION MEMORANDUM

To	Esperanza Blackwood and Meadowgate Terminals Retirement-Plan Committee
From	Personal Financial Advisor, with Tanneryside Solutions review
Subject	Measurement design for one-on-one retirement advice sessions before the November renewal decision
Decision requested	Approve one bounded evaluation design and its access safeguard; no employee result, cost, response rate, or baseline is asserted in this memorandum.

Recommendation

Approve a pre-specified, plant-aware randomized invitation with a 90-day delayed invitation for the comparison group. Randomize invitations contemporaneously within protected blocks, estimate the intention-to-treat effect of being invited, and give every eligible employee access after the measurement window. Permit immediate access only through the documented serious-urgency pathway below. This is the single recommended design because it separates advice from self-selection and concurrent plan changes while preserving eventual access in a 340-employee workforce.

Context and decision frame

Meadowgate has Ohio and Ontario plants with different wage bases, union arrangements, and employer-match formulas. Tanneryside Solutions funds the advice sessions. Last year, deferral changes coincided with auto-escalation; timing alone cannot attribute those changes to advice. Plan changes also occur on different schedules. The committee therefore needs a contemporaneous, invitation-based comparison, not an attendee/nonattendee or simple before/after story.

All numerical items below are design assumptions for committee approval or quantities to estimate from Meadowgate records. They are not claims about employee behavior, costs, participation, baseline deferrals, or prior results.

1. Selected design: who is compared and when

Element	Locked rule for the evaluation
Eligible population	Active employees at either plant who are plan-eligible and have a regular-payroll record at the assignment-file freeze. Exclude only employees already scheduled to separate before the 90-day follow-up, those legally or contractually ineligible for the offered service, and approved urgent-access cases; log every reason. Do not redefine eligibility after outcomes are seen.
Assignment / comparison	Before outcome review, assign a unique study ID and randomly invite roughly half of eligible employees now and the remainder after 90 days. Use constrained randomization within plant, union status, pre-invitation deferral band, and pre-specified wage band; retain the two match formulas as a plant/formula field. If a cell is too sparse for a stand-alone split, use the pre-specified minimization rule across these fields rather than an unbalanced micro-split.
Primary estimand	The adjusted difference in mean change in employee elective deferral rate, in percentage points, for employees assigned an immediate invitation versus employees assigned delayed invitation. This is an offer (intention-to-treat) effect, not the effect among people who attend.
Observation periods	T-2 to T0: the two completed regular payrolls before the invitation is sent (baseline). T0: locked assignment and invitation date. T0 to T+21 calendar days: immediate-group scheduling/session window. T+22 to T+35: completion window. T+60 to T+90: outcome payroll window, using the last two completed regular payrolls ending in that interval. The delayed group receives its invitation after T+90.
Access and crossover	Comparison employees may use ordinary plan tools but are asked not to use the Tanneryside channel until their scheduled invitation. Provider records any early booking, outside-advice disclosure, or crossover. Primary analysis keeps original assignment regardless of attendance or crossover; a secondary exposure description quantifies contamination without recasting it as causal proof.

Why this is credible with limited headcount

Blocking protects the major structural differences rather than pooling Ohio and Ontario as interchangeable. The constrained allocation checks representation before invitations go out; it improves balance but leaves fewer degrees of freedom than a large trial. The analysis is deliberately parsimonious: assignment plus pre-specified plant/formula, union, baseline deferral, wage-band, and dated plan-change terms, with no data-driven subgroup search. The primary report is pooled only after these adjustments; Ohio and Ontario estimates are pre-specified descriptive checks, not separate renewal verdicts.

2. A humane, narrow urgent-access rule

Immediate advice outside the assigned timing is permitted only when a benefits/HR triage owner documents an objectively time-sensitive, seriously consequential financial circumstance: for example, an imminent loss of housing, active insolvency or collection action, a pending court or benefit-election deadline, or an urgent retirement/disability decision with a documented deadline. A wish to act sooner, preference for a particular appointment, general financial anxiety, or ordinary convenience is not sufficient. The triage owner records only the minimum necessary urgency category, verification date, eligibility, assigned arm, access date, and deviation code; detailed circumstances stay outside the analytic file. These cases receive help without delay, remain visible in a confidential deviation log, and are reported separately from the primary eligible population; a sensitivity analysis includes them by original assignment where records permit.

3. Outcomes, dates, and analysis record

Metric hierarchy	Definition and use
Primary behavioral outcome	For each eligible employee, employee elective deferral rate = employee pre-tax plus Roth elective deferral dollars / eligible compensation dollars, measured over the defined payroll window. Primary change = outcome-window rate minus baseline-window rate, in percentage points. Employer match dollars are excluded: match is a separate employer contribution, not employee saving behavior.
Secondary implementation outcomes	Invitation delivered; scheduling; attendance; session completion by T+35; and timely provider file receipt. These show reach and delivery quality, not causal impact. Attendance may support a qualified exploratory complier analysis only if reported separately from the invitation effect.
Supporting financial views	Report the adjusted absolute dollar change in employee elective deferrals and the percentage-point change in rate; report the relative change in rate only when the record-derived baseline is nonzero. Rate-based reporting protects against wage/pay changes being mistaken for saving behavior. Do not combine match and employee dollars.
Concurrent changes	For each participant and payroll, retain auto-escalation eligibility, election, and effective date; retain every plan amendment, match-formula effective date, payroll calendar change, and enrollment event. Include pre-specified dated indicators and run a sensitivity analysis excluding windows crossed by a material effective date. No common intervention date is assumed.
Decision record	Before T0, the committee-approved analyst stores a dated protocol, eligibility snapshot, random seed/allocation file, outcome code, plan-change calendar, threshold, missing-data rule, and subgroup hierarchy in a restricted decision record. Baseline balance is checked on recorded pre-invitation fields only; any imbalance is documented, not repaired using outcome data.

Analysis and missing records

The primary model follows original invitation assignment for everyone in the locked eligible population, including invited employees who never attend. It adjusts only for the listed pre-specified fields and date indicators; wage enters through rate normalization and wage band, while match formula is retained as a plant/formula effect modifier. Missing payroll follow-up is missing, not zero change. Report its count and reason by arm, use available-record analysis with an inverse-probability sensitivity analysis based only on baseline fields, and show a bounded worst-case sensitivity. Transfers retain original assignment and plant label with a transfer flag; terminations, leave, and new hires are reported as censored/late entrants under the locked rule, never silently dropped. Deferral changes before T0 belong only to baseline; post-invitation changes belong only to follow-up.

4. Decision rule and interpretable results

Committee rule	Meaning
Minimum effect to approve	Design assumption requiring approval: +0.50 percentage points in the primary adjusted invitation effect on employee elective deferral-rate change at T+60 to T+90.
Evidence of benefit	Renew funding if the point estimate is positive and the two-sided 95% confidence interval lower bound is at least +0.50 percentage points, subject to data-integrity review.
Too imprecise to decide	Do not infer no effect if the interval crosses +0.50 percentage points. The result is inconclusive; it says this bounded study cannot determine whether the practically meaningful invitation effect was achieved.
No practically meaningful benefit	Treat the evidence as against renewal on behavioral-impact grounds if the two-sided 95% confidence interval upper bound is below +0.50 percentage points. This does not prove advice has no value for any individual or outcome.
Multiplicity / subgroups	The primary outcome and threshold decide. Completion, attendance, dollars, and Ohio/Ontario descriptive estimates cannot override it. Plant estimates are descriptive because the workforce is small; no favorable post hoc subgroup may create an overall conclusion.

Intervals communicate precision, not a mechanical truth test. The committee should read a null point estimate with its interval, data completeness, crossover, and documented plan-change exposure. A low completion rate is an operational finding; it is not evidence that the offer did or did not change retirement saving.

5. Data, privacy, and accountable owners

Owner	Required record / control
Plan administrator	Locked eligibility, employment/plant/union flags, payroll dates, eligible compensation, employee pre-tax and Roth deferrals, employer match, wage/pay changes, transfer/termination/leave/new-hire status, enrollment and auto-escalation dates, plan-amendment and match-formula effective dates. Generates the internal study ID and retains the identity crosswalk.
Tanneryside Solutions	Study ID only: invitation sent/delivered, scheduling, attendance, completion, session date/time, early-contact/crossover flag, and minimum deviation code. No session content enters the analytic file.
Independent analyst	Receives de-identified study-ID analytic file; maintains protocol, code, balance log, missingness report, and reproducible output. May not access the identity crosswalk or detailed urgency documentation.
HR/benefits triage owner	Applies and documents the narrow urgency rule; may see identity and verification evidence but sends only the minimum deviation fields to analysis.
Committee / Esperanza Blackwood	Approves the protocol, threshold, access boundary, and renewal interpretation; receives aggregate results only. The plan administrator and provider revoke project access after the final committee report, retain the crosswalk and protocol only under Meadowgate retention policy, and certify destruction/return of analytic extracts.

Privacy and small populations

Direct identifiers remain with the plan administrator. Analytic extracts use a stable internal study ID, restricted role-based access, encrypted transfer, and no session narrative. Reports suppress any plant, union, or other subgroup cell smaller than the committee-approved minimum-cell rule; where suppression would reveal a complement, aggregate or withhold both. No individual-level result, urgency reason, or identifiable small cell is delivered to Tanneryside or the committee.

6. Alternatives considered and rejected

Alternative	Why it is not the selected decision
Attendee vs. nonattendee	Attendance is chosen by employees and may reflect motivation, financial stress, schedule, or wages. Adjustment cannot guarantee removal of unrecorded self-selection; it is not the primary causal estimate.
Simple before / after	It would repeat last year's attribution problem: auto-escalation and staggered plan changes can move deferrals independently of advice. Without a contemporaneous comparison, a null or increase is not interpretable.
Nonrandom phased rollout by plant	Operationally simple, but plant is inseparable from union conditions, wage base, and match formula. With only two plants, plant timing is a weak counterfactual.
Immediate universal access, observational adjustment	Maximizes convenience but removes the only contemporaneous assignment-based comparison. It may still monitor service delivery, but cannot support the requested funding attribution.

Final decision

Approve the plant-aware randomized invitation with a 90-day delayed invitation, the documented serious-urgency access pathway, and the +0.50 percentage-point primary invitation-effect rule. This is the committee’s selected basis for deciding whether to renew the Tanneryside Solutions advice-session budget in November.